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NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***Fourth Semester B.Tech (Minor) (CBCS) Degree Examinations**

May 2024

CS2005-1- INTRODUCTION TO OBJECT ORIENTED PROGRAMMING

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What is the primary purpose of constructors in Java?**
A) To initialize object variables B) To declare class variables
C) To define object methods D) To implement method overloading
2. **When declaring objects in Java, what is the primary purpose of the 'new' keyword?**
A) To declare a new class B) To allocate memory for an object
C) To define object methods D) To access class variables
3. **Which keyword is used to refer to the current object within a class method?**
A) this B) super
C) final D) static
4. **In Java, which keyword is used to restrict access to class members within the same package?**
A) public B) private
C) protected D) static
5. **What does method overloading in Java allow?**
A) Defining multiple methods with the same name but different parameters B) Redefining a method with a different return type
C) Defining a method with the same name and parameters in different classes D) Hiding methods from other classes
6. **How does Java handle multiple inheritance?**
A) Java supports multiple inheritance through interfaces B) Java does not support multiple inheritance
C) Java allows multiple inheritance only through abstract classes D) Java allows multiple inheritance through the 'extends' keyword
7. **What is the primary purpose of using the 'super' keyword in Java inheritance?**
A) To call superclass methods from within a subclass B) To create instances of subclasses
C) To prevent method overriding D) To declare abstract methods
8. **When is it appropriate to use the 'final' keyword with inheritance in Java?**
A) When defining a superclass B) When declaring a class as abstract
C) When preventing a class from being extended D) When overriding methods in a subclass
9. **What does it mean for a class to be declared as abstract in Java?**
A) It cannot have any constructors B) It must implement all of its methods
C) It cannot be instantiated directly D) It must extend another abstract class
10. **What is a multilevel hierarchy in Java inheritance?**
A) A hierarchy with multiple levels of inheritance depth B) A hierarchy with only one level of inheritance
C) A hierarchy where subclasses inherit from more than one superclass D) A hierarchy where subclasses cannot inherit from each other

11. In Java, what is the purpose of using packages?
 - A) To group related classes and interfaces
 - B) To define methods
 - C) To restrict access to class members
 - D) To create objects
12. How are classes organized within packages in Java?
 - A) By using the 'class' keyword followed by the package name
 - B) By placing classes in separate directories corresponding to their package names
 - C) By importing classes into packages
 - D) By using the 'package' keyword followed by the class name
13. Can you import all classes from a package in Java with a single statement?
 - A) Yes, using the * wildcard character
 - B) No, each class must be imported individually
 - C) Yes, by using the 'import all' keyword
 - D) No, Java does not support importing all classes from a package
14. What is the purpose of access protection in Java?
 - A) To prevent classes from being imported
 - B) To restrict access to certain members of a class
 - C) To define interfaces
 - D) To specify package names
15. In Java, which type of exception must be caught or declared in the method signature?
 - A) Checked exceptions
 - B) Unchecked exceptions
 - C) RuntimeExceptions
 - D) IOExceptions
16. What is the purpose of using multiple catch clauses in Java?
 - A) To catch exceptions of different types
 - B) To catch all exceptions with a single catch block
 - C) To re-throw exceptions to the calling method
 - D) To specify which exceptions should be thrown
17. How does Java handle uncaught exceptions?
 - A) By ignoring them
 - B) By printing the exception stack trace and terminating the program
 - C) By automatically handling them
 - D) By passing them to the 'finally' block
18. What is the primary purpose of exception handling in Java?
 - A) To create errors intentionally
 - B) To ignore errors during program execution
 - C) To gracefully handle runtime errors
 - D) To terminate the program immediately
19. What is the purpose of the indexOf() method in Java string handling?
 - A) It returns the index of the first occurrence of a specified substring
 - B) It checks if a string contains a specified character
 - C) It replaces occurrences of a substring with another substring
 - D) It removes leading and trailing whitespace characters
20. What does the trim() method do in Java string handling?
 - A) Removes all leading and trailing whitespace characters
 - B) Converts the string to uppercase
 - C) Splits the string into substrings
 - D) Returns the substring starting from the specified index

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the difference between a constructor and a method in Java. How are they invoked? Provide code examples to support your explanation.	6	L*2	1	1
	b) Design a Java class representing a Book. Include attributes such as title, author, and price. Implement methods to set and get these attributes. Provide a sample usage of this class in another Java program.	6	L3	1	2
	c) Discuss the significance of the "this" keyword in Java. Provide scenarios where its usage is beneficial.	4	L2	1	1
2.	a) Discuss the concept of static and final keywords in Java. Provide examples illustrating their usage.	6	L2	1	1
	b) Develop a Java program that utilizes method overloading to calculate the area of different geometric shapes such as rectangle, circle, and triangle. Provide a step-by-step explanation of your code.	6	L3	1	2

c)	Develop a Java program that utilizes command line arguments to calculate the average of a set of numbers	4	L2	1	2
3. a)	Explain the concept of method overriding in Java. Provide an example demonstrating method overriding along with a brief explanation of the process.	6	L2	2	1
b)	What is inheritance? Develop a Java program that demonstrates the creation of a multilevel hierarchy using inheritance. Define classes such as Animal, Mammal, and Dog, each inheriting from the previous class. Include appropriate methods and attributes.	6	L3	2	2
c)	Discuss the significance of using the super keyword in Java inheritance. Provide examples to illustrate its usage	4	L2	2	1

Unit – II

4. a)	Define what interfaces are in Java. Discuss their role in achieving multiple inheritance. Give example for declaration and implementation of interfaces	8	L2	3	1
b)	Discuss the following concepts in interfaces with a small example i. Extending interfaces ii. Default interfaces	8	L1	3	1
5. a)	Define what packages are in Java. Explain their significance in organizing code. Provide an example for the creation and usage of packages.	6	L2	3	1
b)	Explain the purpose of the finally block in Java exception handling. How does it differ from the catch block? Provide examples	6	L2	3	1
c)	Explain the process of importing packages in Java. How does it simplify code development? Demonstrate the use of the import statement with example.	4	L2	3	1
6. a)	Explain the purpose of the try and catch blocks in Java exception handling. How do they work together to handle exceptions? Give example	8	L2	4	1
b)	Discuss any six most commonly occurring built-in exceptions in Java.	8	L2	4	1

Unit – III

7. a)	What are the different ways to declare and initialize the Strings in Java? Why Strings are called immutable?	8	L2	5	1
b)	Explain the purpose and usage of basic string methods in Java. Discuss methods such as length() , charAt() , substring() , and indexOf() with example.	8	L2	5	1
8. a)	Define what file handling is in Java. Give example to illustrate file handling operations	8	L2	5	1
b)	Write a Java program to create a serial access file named "EmployeeData.txt" and add records of employees including their name, age, and salary.	8	L3	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
